Running pipelines HW1. Kulikova E.D.

Firstly, Nextflow was installed via provided instructions:

```
curl -s https://get.nextflow.io | bash
chmod +x nextflow
mv nextflow $HOME/.local/bin/
```

rnaseq-nf pipeline was copied from Git

Test data running

```
nextflow run rnaseq-nf -with-docker
nextflow run rnaseq-nf -with-conda --outdir 'results_conda'
```

```
(base) rina@LAPTOP-0K822FEP:/mnt/c/Users/Admin/Downloads$ nextflow run rnaseq-nf -with-conda --outdir 'results_conda'

 N E X T F L O W  ~  version 25.10.4

Launching `rnaseq-nf/main.nf` [hungry_wilson] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /mnt/c/Users/Admin/Downloads/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /mnt/c/Users/Admin/Downloads/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results_conda

[-        ] RNASEQ:INDEX  -
[-        ] RNASEQ:FASTQC -
[-        ] RNASEQ:INDEX  -
[-        ] RNASEQ:FASTQC -
executor >  local (3)
executor >  local (4)
[d8/68ee46] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[2c/de4c93] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✔
[e7/5fb54e] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✔
[a9/96ed79] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> results_conda/multiqc_report.html
```

Report for test data is in the folder “multiqc_test.html”.

Custom files running

I used reads from a GEO dataset GSE110004.

```
wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR635/008/SRR6357070/SRR6357078_1.fastq.gz
wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR635/008/SRR6357070/SRR6357078_2.fastq.gz
nextflow run rnaseq-nf -with-docker \
--outdir 'data_QC/results' \
--reads 'data_QC/reads/SRR6357070_{1,2}.fastq.gz' \
--transcriptome '/mnt/c/Users/Admin/Downloads/data_QC/ref/genome.fasta'
```

```
(base) rina@LAPTOP-0K822FEP:/mnt/c/Users/Admin/Downloads$ nextflow run rnaseq-nf -with-docker --outdir 'data_QC/results' --reads 'data_QC/reads/SRR6
357070_{1,2}.fastq.gz' --transcriptome '/mnt/c/Users/Admin/Downloads/data_QC/ref/genome.fasta'

N E X T F L O W  ~  version 25.10.4

Launching `rnaseq-nf/main.nf` [sad_babbage] DSL2 - revision: 295b10c8d2

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /mnt/c/Users/Admin/Downloads/data_QC/ref/genome.fasta
reads        : data_QC/reads/SRR6357070_{1,2}.fastq.gz
outdir       : data_QC/results

executor >  local (4)
[12/2173aa] RNASEQ:INDEX (genome)      [100%] 1 of 1 ✔
[6b/83d530] RNASEQ:FASTQC (SRR6357070) [100%] 1 of 1 ✔
[ac/8a6f83] RNASEQ:QUANT (SRR6357070)  [100%] 1 of 1 ✔
[fe/40671d] MULTIQC                    [100%] 1 of 1 ✔

Done! Open the following report in your browser --> data_QC/results/multiqc_report.html
```

Report for custom data is in the repo “multiqc_SRR6357070.html”